TRANSIT-BASED EXOPLANET DETECTION AND HABITABILITY ASSESSMENT OF THE STAC SYSTEMS

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Abstract—This report analyzes relative-brightness observations of three stars, STAC-1, STAC-2, and STAC-3, to identify candidate transiting exoplanets and compare their sizes, orbital distances, and approximate habitable-zone locations. Transit depths are used to estimate planetary radii, while orbital periods and stellar masses are used to estimate orbital distances. The report then compares the three candidate systems and identifies a target for further habitability study. The analysis uses simplified assumptions, including circular orbits and central transits, and does not account for limb darkening, stellar flares, or measurement uncertainty.

Index Terms—exoplanets, transit photometry, orbital period, planetary radius, habitable zone.

INTRODUCTION

This report is to analyse the relative-brightness observation of three stars – STAC-1, STAC-2 and STAC-3, and their stellar properties to detect possible exoplanets and select one planet for further habitability study.

EQUATIONS AND METHODS

Property	STAC-1	STAC-2	STAC-3
Mass	1.00 M⊙	0.80M⊙	0.50M⊙
Radius	1.00 R⊙	0.80R⊙	0.50R⊙
Luminosity	1.00 L⊙	0.40L⊙	0.08L⊙
Surface Temperature	5800 K	5000 K	3800 K

STELLAR PROPERTIES

ASSUMPTIONS

Orbits are circular

Transits are central

Limb darkening, flares and measurement uncertainty are ignored

EQUATIONS

Using the given information transit depth is estimated from

$$\delta = (F_{normal} - F_{transit}) / F_{normal} \quad \ldots 1$$

The assignment gives the approximation

$$\delta \approx (R_p/R_s)^2,$$

so the planet radius is

$$R_p \approx R_s\sqrt{\delta} \quad \ldots 2$$

For orbital distance, the supplied relation is

$$P^2 = a^3/M,$$

with P in years, a in AU, and stellar mass M in solar masses. Rearranging gives

$$a = (P^2M)^{(1/3)} \quad \ldots 3$$

Periods are converted from days to years using 365.25 days per year. Distances are converted using

1 AU = 149.6 million km.

The approximate habitable-zone distance is calculated from

$$d_{HZ} \approx \sqrt{(L/L\odot)}\ AU \quad \ldots 4$$

PART A: DETECTING THE EXOPLANETS

IDENTIFYING THE TRANSITS :

The transit method detects a possible exoplanet by measuring a small, repeated decrease in a star's observed brightness when a planet passes between the observer and its star As the normal brightness is 1.0000, lowering of the brightness can be a transit.

STAC-1 :

Dips occur at interval close to 4 days, at days 3, 8, 12, 16, 20 with brightness constantly falling to 0.9970.

STAC-2 :

Dips occur at interval of 40 days, at days 20, 60, 100 and 120 with brightness constantly falling to 0.9990.

STAC-3:

Dips occur at an interval of 40 and 30 days, occurring on days 20, 60, 100, 140 and 170 with inconstant brightness (0.9990, 0.992, 0.9991, 0.9993, and 0.9990). This inconstant brightness and the day 170 dip does not fit the pattern and should be treated cautiously.

As all the three stars are showing periodic decrease in brightness so each looks like it hosts a transiting planet.

ORBITAL PERIODS :

Period is the time between one transit and the next. To find the orbital periods we will use the span of first and the last transit and divided by the number of cycles between them.

For STAC-1 :

$P_1 = (20-3)/4 = 4.25d$

For STAC-2:

$P_2 = (180-20)/2 = 80d$

For STAC-3:

The first four dips are spaced neatly 40 d apart, so at first glance the period looks like 40 d. But then day 180, where the next dip should be, is perfectly normal, while an extra dip shows up at day 170. A period of exactly 40 d cannot explain that.

For the brightness, the first instinct is to consider noise or starspots which could've been it but these cannot explain irregular dip at day 170.

The first four dips are spaced neatly 40 d apart, so at first glance the period looks like 40 d. But then day 180, where the next dip should be, is perfectly normal, while an extra dip shows up at day 170. A period of exactly 40 d cannot explain that.

Another explanation is that the true period is a little shorter than 40 days. Every time we look 40 days later, the planet has finished one orbit and then some, so it sits slightly further along its orbit than before. The transit slowly drifts relative to our fixed sampling times, and after a few cycles it slides out of the sampled window entirely. This also fits the slowly shrinking depths (0.0010, 0.0008, 0.0009, 0.0007), since a planet crossing near the edge of the disc blocks less light.

Hence,

$P_3 = (170-20)/4 = 37.5d$

3. TRANSIT DEPTHS:

From Eq. (1):

STAC-1: δ_1 - (1.0000 - 0.9970)/1.0000 0.0030 (0.30%).

STAC-2: δ_2 - 0.0010 (0.10%).

STAC-3: the individual depths are 0.0010, 0.0008, 0.0009, 0.0007, 0.0010, with mean

δ_3 - 0.00088 ≈ 0.0009 (0.09%).

4. WHY A PERIODIC DECREASE INDICATE AN EXOPLANET:

A revolution of planet is periodic. Hence, periodic decrease in stellar brightness of equal transit depths (as it is the same planet each time) strongly indicates a planet. Other effects, such as starspots or random flickering, rarely repeat with a fixed period and a constant depth.

PART B: PLANET SIZE

First we convert the stellar radii to kilometres: 1.00 x 696,000 =696,000, 0.80 x 696,000 = 556,800 and 0.50 x 696,000 =348,000 km for STAC-1, -2 and -3 respectively. Then Eq. (2) gives

$R_{p,1}$ = 696,000/0.0030 = 696,000 × 0.05477 ≈ 38,100 km,

$R_{p,2}$ =556,800/0.0010 = 556,800 x 0.03162 ≈ 17,600 km,

$R_{p,3}$ = 348,000/0.0009 = 348,000 x 0.0300 ≈ 10,440 km.

Dividing by R⊕ = 6,371 km turns these into 5.98, 2.76 and 1.64 R⊕.

Table gives the results with the size classes. Even if we use the full spread of STAC-3 depths (0.0007 to 0.0010), the radius only ranges from 1.45 to 1.73 Rp-so STAC-3 planet stays a Super-Earth either way.

Star of planet	δ	RP(km)	Rp/R⊕	Class
STAC-1 b	0.0030	38100	5.98	Neptune-sized
STAC-2 b	0.0010	17600	2.76	Super-Earth/ Mini Neptune
STAC-3 b	0.0009	10440	1.64	Super-Earth

PART C: PLANETARY ORBITS

Using Eq. (3), (also converting period from days to years by dividing it by 365.25)

STAC-1 b (M=1.00): P = 4.25/365.25 = 0.011636 yr,

$P^2 = 1.354 \times 10^{-4}$, $a^3 = 1.00 \times 1.354 \times 10^{-4}$, so a = $\sqrt[3]{1.354 \times 10^{-4}}$ ≈ 0.0514 AU = 0.0514 × 149.6 × 10^6 ≈ 7.7 × 10^6 km. This is about 19 times closer to its star than Earth is to the Sun.

STAC-2 b (M - 0.80): P = 80/365.25 = 0.21903 yr, P^2 =0.047974, a^3 = 0.80 × 0.047974 = 0.038379, so a ≈ 0.337 AU ≈ 5.05 × 10^7 km, about 3 times closer than Earth (similar to Mercury's 0.39 AU).

STAC-3 b (M - 0.50): P = 375/365.25 = 0.10267 yr. P^2 = 0.010541, a^3 = 0.50 x 0.010541 = 0.0052706, so a ≈ 0.174 AU ≈ 2.60 × 10^7 km, about 5.7 times closer than Earth.

PART D: MOST PROMISING WORLD

HABITABLE ZONE COMPARISON

The simplified habitable-zone reference distances are:

STAC-1: d_{HZ} ≈ √1.00 = 1.00 AU. Its estimated orbit is 0.049 AU, far inside this reference distance.

STAC-2: d_{HZ} ≈ √0.40 ≈ 0.632 AU. Its estimated orbit is 0.337 AU, inside the reference distance.

STAC-3: d_{HZ} ≈ √0.08 ≈ 0.283 AU. Its estimated orbit is 0.182 AU, also inside the reference distance, but it is the closest of the three in relative terms: 0.182/0.283 ≈ 0.64, compared with 0.337/0.632 ≈ 0.53 for STAC-2 and 0.049/1.00 ≈ 0.05 for STAC-1.

THE CHOSEN ONE

We choose STAC-3 planet for the following reasons:

It is closest to the habitable zone as compared to 1 and 2.

It is a small planet (super-earth sized) which indicates it most likely has a rocky composition.

It's star is the least luminous so being close to it doesn't causes extreme heating.

Summary:

By the analysis STAC-3 is the best of the three choices but still not the perfect one. It is nearest to the habitable zone but a little too close to the star making it a little too hot. Also habitability depends on other factors such as atmosphere, magnetic field and water. The summary of our observation is stated in the table below.

System	Period	Depth	Radius (R⊕)	Class	Orbit(AU)	Orbit(million km)	dHZ(AU)
STAC-1	4.25d	0.0030	5.98	Neptune-sized	0.051	7.7	1.000
STAC-2	80d	0.0010	2.76	Super-Earth /Mini-Neptune	0.337	50.5	0.632
STAC-3	37.5d	0.0009	1.64	Super - Earth	0.174	26	0.283